

Atmospheric Chemistry Modeling in Highly Constrained Ephemeral Environments: Feasibility Assessment for the Delhi NCR Domain

1. Introduction to the Computational Challenge

The deployment of a high-resolution, operational chemical transport model (CTM) within a continuous integration and continuous deployment (CI/CD) pipeline presents a formidable computational paradigm. This report exhaustively assesses the mathematical, architectural, and physical feasibility of automating a 72-hour air quality forecasting system over the Delhi National Capital Region (NCR) domain, constrained strictly within a standard GitHub Actions runner. The domain, spanning approximately 27.0–29.9°N and 75.8–78.1°E, represents one of the most chemically complex atmospheres globally, characterized by intense anthropogenic multiphase emissions, severe wintertime subsidence inversions, and highly non-linear secondary organic aerosol (SOA) formation regimes.

The architectural constraints imposed by the GitHub Actions runner environment define strict, unyielding boundaries for the numerical simulation. The environment provides four virtual Central Processing Units (vCPUs), symmetrical multiprocessing without Message Passing Interface (MPI) clustering capabilities, 16 gigabytes (GB) of Random Access Memory (RAM), and ephemeral storage that is purged upon job completion. Crucially, the environment enforces a rigid six-hour wall-clock job termination limit. Because the storage is ephemeral, all meteorological driver data, boundary conditions, anthropogenic emissions inventories, and model source codes must be fetched, compiled (if necessary), interpolated, and executed within this six-hour window.

To generate a scientifically defensible prognostic forecast, an atmospheric chemistry model must numerically integrate the Navier-Stokes equations for fluid dynamics alongside continuity equations for dozens to hundreds of reactive chemical tracers. This report evaluates the viability of executing full three-dimensional Eulerian models (WRF-Chem, CMAQ, CAMx), lighter zero-dimensional and one-dimensional alternatives (PyBox, AtChem2, BOXMOX, PySDM, WRF-SCM), and the integration of open-source high-resolution anthropogenic emissions inventories. Each modeling architecture is analyzed against the specific hardware constraints, concluding with a ranking based on the scientific claim-gained per engineering-day, and an assessment of the minimal cloud architecture required to bypass these constraints should the CI/CD environment prove computationally infeasible.

2. Assessment of Three-Dimensional Eulerian Chemical Transport Models

The primary computational bottleneck for three-dimensional Eulerian atmospheric chemistry

models resides in the dimensionality of the state vector. The simulation must resolve horizontal and vertical advection, turbulent boundary layer mixing, and the extremely stiff ordinary differential equations (ODEs) that govern chemical kinetics.

2.1 WRF-Chem (Weather Research and Forecasting Model Coupled with Chemistry)

WRF-Chem is a state-of-the-science, fully coupled, "online" atmospheric model¹. Because the meteorological and chemical states are integrated simultaneously within the same timestep and on the same grid, it avoids the temporal interpolation errors inherent in offline models. However, this online coupling drastically increases the memory and computational burden per integration step.

To generate a credible 72-hour forecast for the Delhi NCR domain without relying on massive outer domains, the simulation must minimize grid volume while maintaining numerical stability. A single small nested domain is inherently problematic due to boundary condition corruption from the coarse global driver data; however, defining the absolute minimum credible configuration yields a specific set of parameters. The spatial resolution must be restricted to a 10-kilometer horizontal grid spacing. At 10 kilometers, the 320 by 230 kilometer Delhi domain translates to an active grid of 32 by 23 cells. Adding five halo cells on each boundary for numerical relaxation and stability results in a final computational grid of 42 by 33 cells. Vertically, the model requires a minimum of 35 terrain-following eta levels, tightly packed within the planetary boundary layer to accurately capture the shallow nocturnal inversion layers that govern Delhi's wintertime particulate matter accumulation.

The meteorological time step is dictated by the Courant–Friedrichs–Lewy (CFL) stability criterion. For a 10-kilometer grid, the integration step is approximately 60 seconds, meaning a 72-hour prognostic forecast requires 4,320 discrete integration steps. For the chemistry mechanism, simulating Delhi's smog requires the MOZART-4 (Model for Ozone and Related chemical Tracers) gas-phase mechanism coupled with the MOSAIC (Model for Simulating Aerosol Interactions and Chemistry) aerosol module. The Kinetic PreProcessor (KPP) is typically utilized to generate the Fortran code required to solve the highly non-linear, stiff chemical ODEs associated with the MOZART mechanism².

The GitHub Actions environment is fundamentally hostile to this configuration. First, WRF-Chem must be compiled from source, as pre-compiled binaries for specific chemical mechanisms are rarely universally compatible with CI/CD virtual machine architectures. Compiling WRF-Chem with KPP is notoriously memory-intensive. Utilizing multiple threads for compilation routines regularly causes extreme memory allocation spikes that can freeze operating systems equipped with 64 GB of modern DDR5 RAM². To compile successfully within the 16 GB constraint, the build process must be restricted to a single thread, which consumes up to four hours of wall-clock time, effectively exhausting the six-hour job limit before numerical integration even commences².

Assuming a pre-compiled binary could be cached, the runtime constraints remain insurmountable. The memory footprint of WRF-Chem scales linearly with the grid volume and the number of tracked chemical species. For a 42 by 33 by 35 grid tracking approximately 150 prognostic variables, the baseline memory allocation is theoretically between 4 and 6 GB.

However, intermediate arrays utilized by the KPP chemical solver, alongside the overhead required to simulate distributed memory via OpenMP on the four available vCPUs, push peak memory consumption toward 10 to 12 GB. Because the GitHub Actions runner lacks persistent swap space, any memory fragmentation during the execution of the chemical solver will trigger a sudden out-of-memory kernel termination³.

Regarding execution speed, historical benchmarking of WRF-Chem demonstrates that the chemistry solver consumes upward of 80 percent of the total central processing unit cycles. A benchmark of WRF-Chem running a 72-hour forecast on four cores indicates that the numerical integration for a domain of this size, utilizing complex MOZART/MOSAIC chemistry, requires between 12 and 18 hours of continuous computation³. This completely violates the six-hour CI/CD job limit. Furthermore, initializing the model requires fetching roughly 2 GB of Global Forecast System (GFS) boundary conditions and processing them through the WRF Preprocessing System (WPS), adding significant I/O overhead.

Under the prescribed constraints, WRF-Chem is explicitly not feasible. The pipeline will break first at the compilation stage due to RAM exhaustion, and if circumvented, will definitively break at the six-hour wall-clock termination limit during the main executable integration.

2.2 CMAQ and CAMx (Offline Chemical Transport Models)

The Community Multiscale Air Quality (CMAQ) model and the Comprehensive Air Quality Model with Extensions (CAMx) represent the "offline" CTM paradigm. These models do not solve the fluid dynamics of the atmosphere; instead, they ingest pre-calculated meteorological fields generated by models like WRF or the National Oceanic and Atmospheric Administration's Global Forecast System (GFSv16)⁷. CMAQ and CAMx read these fields at regular intervals to drive the transport, dispersion, and chemical transformation of pollutants⁹.

The minimum credible configuration for CAMx or CMAQ mirrors that of WRF-Chem: a 42 by 33 horizontal grid at 10-kilometer spacing with 35 vertical levels. The Carbon Bond 6 (CB6) gas-phase mechanism coupled with the AERO7 aerosol module provides the necessary chemical fidelity to model Delhi's secondary organic aerosol precursors. The execution pipeline involves downloading 72 hours of meteorological data, interpolating the fields via the Meteorology-Chemistry Interface Processor (MCIP), processing anthropogenic data through the Sparse Matrix Operator Kernel Emissions (SMOKE) model, and finally running the CMAQ or CAMx executable⁸.

While offline models boast optimized, fast chemical solvers that can theoretically execute slightly faster than an online WRF-Chem run⁹, the offline paradigm introduces an insurmountable Input/Output (I/O) and storage bottleneck within an ephemeral runner. To drive a 72-hour forecast, CMAQ requires hourly three-dimensional meteorological fields⁸. Fetching 72 hours of high-resolution GFSv16 data or pre-computed WRF outputs equates to downloading between 5 and 15 GB of compressed NetCDF or GRIB2 files. Decompressing these files and processing them spatially and temporally through MCIP will immediately exhaust the local ephemeral disk space of standard GitHub Actions runners⁸. Furthermore, reading massive NetCDF arrays into memory for spatial interpolation frequently spikes RAM usage well beyond the 16 GB limit.

Even if the storage constraints were bypassed, the execution of the SMOKE model to generate

gridded, temporally allocated emissions for Delhi is heavily memory and CPU bound¹¹. The sequential pipeline of downloading meteorology, running MCIP, running SMOKE, and executing CAMx or CMAQ for 72 hours on four vCPUs will easily exceed 8 to 10 hours. Consequently, both CMAQ and CAMx are explicitly not feasible. The system will break first at the ephemeral storage limit during meteorological data acquisition, followed rapidly by memory allocation failures during MCIP processing, and finally by exceeding the six-hour wall-clock limit.

3. Lighter Alternatives: Zero-Dimensional and One-Dimensional Models

Given the mathematical and physical impossibility of executing three-dimensional Eulerian CTMs within a 4 vCPU, 16 GB, 6-hour environment, generating a defensible atmospheric chemistry forecast requires reducing the dimensionality of the simulation. By simplifying or entirely removing the fluid dynamic advection schemes, computational resources are liberated, allowing for the execution of highly complex, explicit chemical mechanisms.

3.1 Zero-Dimensional Box Models (AtChem2, BOXMOX, PyBox)

A zero-dimensional (0D) box model operates on the assumption that the atmosphere over the Delhi NCR is a perfectly mixed reactor. By removing horizontal and vertical advection (the Navier-Stokes equations), 99 percent of the computational burden is eliminated. This allows researchers to deploy hyper-complex chemical mechanisms that would paralyze a 3D model. Models such as AtChem2, BOXMOX, and PyBox serve as wrappers for highly sophisticated ODE solvers⁴. BOXMOX, for instance, utilizes the Kinetic PreProcessor (KPP) to translate chemical equations into efficient Fortran code, enabling the rapid integration of complex mechanisms⁴. The computational freedom provided by the 0D architecture allows for the utilization of the Master Chemical Mechanism (MCM)¹². The MCM is a near-explicit mechanism detailing the degradation of volatile organic compounds (VOCs), containing over 17,000 distinct reactions and tracking more than 5,000 intermediate species. Alternatively, highly detailed lumped mechanisms like SAPRC-99 or CB6 can be utilized¹⁴.

A 72-hour forecast using AtChem2 or BOXMOX with a full MCM subset tailored to Delhi's specific VOC emission profile requires less than 200 megabytes of RAM and completes its execution in under five minutes on a single vCPU. Therefore, 0D models fit effortlessly within the 16 GB and six-hour constraints. However, the first point of failure for a 0D model is its physical realism. A box model lacks a representation of planetary boundary layer collapse, thermal subsidence, and horizontal transboundary transport. It cannot account for the advection of smoke from agricultural crop-residue burning in upwind states like Punjab and Haryana, which is a critical driver of Delhi's autumn particulate matter peaks.

3.2 One-Dimensional Column and Parcel Models (PySDM, WRF-SCM)

A one-dimensional Single-Column Model (SCM) introduces vertical resolution, allowing for the simulation of boundary layer dynamics, aerosol activation, and vertical turbulent mixing without incurring the massive computational cost of horizontal grids¹⁵.

The Weather Research and Forecasting Single-Column Model (WRF-SCM) offers a lightweight

testing environment that utilizes the exact physical parameterizations of the full 3D WRF model but restricts the domain to a single vertical column (typically a 3 by 3 grid with periodic boundary conditions)¹⁷. A 72-hour forecast utilizing WRF-SCM coupled with complex chemistry requires less than 1 GB of RAM and executes in minutes¹⁸.

Alternatively, PySDM provides a Pythonic, high-performance implementation of the Super-Droplet Method (SDM). The SDM is a particle-based Monte-Carlo algorithm designed for atmospheric aqueous chemistry and aerosol-cloud interactions²⁰. By representing the dispersed aerosol phase as computational "super particles," PySDM overcomes the numerical diffusion and state-space limitations of traditional Eulerian bin methods²¹. PySDM effectively models multiphase chemistry, coagulation, and condensation²². Crucially, its Pythonic architecture bypasses the legacy Fortran compilation nightmares associated with KPP, executing efficiently via Just-In-Time (JIT) compilation on modern CPUs, and fits seamlessly into standard CI/CD environments²⁴. Like WRF-SCM, PySDM easily respects the 2 GB memory and six-hour wall-clock limits, generating rapid, high-fidelity vertical profiles of aerosol chemistry.

3.3 Defensible Scientific Claims vs. Statistical Surrogates and CAMS

If constrained to a one-dimensional column model over Delhi, the critical question becomes: why build a deterministic physical model when one could simply consume coarse global output from the Copernicus Atmosphere Monitoring Service (CAMS) or train a machine learning statistical surrogate?

A statistical surrogate model (such as a Random Forest or Long Short-Term Memory network) trained on historical Delhi air quality data is inherently interpolative. It learns historical correlations between meteorology and surface observations. However, a single-column chemistry model can legitimately claim to predict out-of-distribution, non-linear chemical responses to unprecedented emission perturbations—a claim a statistical surrogate cannot make. For example, if a policy abruptly bans a specific class of vehicular emissions, altering the NO_x to VOC ratio, a statistical surrogate will fail to predict the resultant shift in ozone formation because it has never observed that specific chemical regime in its training data. A 1D column model with an explicit chemical mechanism (like MCM or MOZART) relies on fundamental chemical kinetics rather than historical correlation, rendering its predictions of out-of-distribution emission scenarios scientifically defensible.

Furthermore, a 1D column model provides defensible insights that are mathematically impossible to derive from CAMS global output. CAMS operates at a spatial resolution of roughly 0.4 degrees (approximately 40 kilometers)²⁶. At this coarse resolution, massive point sources and intense urban vehicular emissions in Delhi are artificially diluted across a 1,600 square kilometer grid cell. Because ozone and secondary organic aerosol formation are highly non-linear, artificially diluting NO_x shifts the grid cell from a VOC-limited regime (which characterizes Delhi's dense urban core) to a NO_x -limited regime. A single-column model, initialized with localized high-resolution emissions (e.g., at 1 kilometer resolution) and driven by localized surface heat fluxes, accurately resolves the severe nocturnal boundary layer titration

and diurnal radical partitioning unique to Delhi's specific emission density. Consequently, the 1D model can defensibly claim to represent the true non-linear chemical regime and extreme urban SOA accumulation, which is systematically misrepresented by the artificial dilution inherent in CAMS output.

4. Openly Licensed High-Resolution Emissions Inventories

To drive a 1D column or 0D box model accurately, the simulation requires highly resolved, sector-specific anthropogenic emissions data. An exhaustive evaluation of openly licensed inventories suitable for the Delhi NCR domain at resolutions of 10 kilometers or finer reveals several distinct options.

The following table summarizes the key characteristics, licensing, and CI/CD integration viability of each major anthropogenic emission inventory available for the Indian subcontinent.

Inventory	Resolution	Base Year	Key Sectors Covered	License	Download Mechanism & CI/CD Viability
EDGAR v8.1 [cite: 28]	$0.1^{\circ} \times$ (~10 km)	Up to 2022 (Air Pollutants) ²⁸	Power, Industry, Transport, Residential (solid fuels), Agriculture ²⁸ .	CC BY 4.0 ²⁸	Direct HTTP/FTP via JRC or Zenodo. NetCDF format ²⁸ . Highly viable for automated wget pipelines.
CAMS-GL OB-ANT v6.2 [cite: 27, 31]	$0.1^{\circ} \times$ (~10 km)	2000–2024 (projected) ²⁷	17 sectors including agriculture, residential, and transportation ²⁷ .	Copernicus Open Data ²⁶	API access via Atmosphere Data Store (ADS) ²⁶ . Highly viable , requires API key stored

					as a GitHub Secret.
SMoG-India [cite: 33, 34, 35]	$\leq$ km	2015, updated 2019 ³⁵	Specific Indian profiles: brick kilns, massive biomass burning, residential ³⁵ .	Open (Academic/Research)	Zenodo or Google Sites. Provided in ASCII and NetCDF ³⁸ . Highly viable for automated downloading.
REAS v3.2 [cite: 40, 41]	$0.25^\circ \times$ (~25 km)	1950–2015 ⁴¹	Energy, Industry, Transport, Domestic ⁴⁰ .	Open (Non-commercial)	NIES Japan portal. Requires HTML parsing ⁴² . Less viable due to outdated base year and brittle scraping requirements.
SAFAR-Delhi [cite: 44, 45]	400 m – 1 km	2018 ⁴⁵	Street-level transport, municipal waste burning, localized point sources ⁴⁵ .	Restricted / Not Open	Requires institutional requests to the Ministry of Earth Sciences ⁴⁵ . Not viable for frictionless, automated programmatic CI/CD pipelines.

For the specific application of modeling Delhi's unique chemical atmosphere, the SMOG-India inventory provides the highest scientific value due to its inclusion of highly disaggregated residential solid fuel use and regional biomass burning profiles developed directly by the Indian Institute of Technology, Bombay³³. When combined with the global baseline of EDGAR v8.1, researchers can construct a highly accurate localized emission forcing dataset that is easily fetched into an ephemeral GitHub Actions environment via standard secure shell commands²⁸.

5. Return on Investment: Claim-Gained per Engineering-Day

Given the computational reality of the GitHub Actions runner, the engineering effort required to force a model into compliance must be carefully weighed against the scientific validity of the resulting output. An architecture that consumes weeks of engineering time to compile, only to produce scientifically flawed output due to extreme resolution compromises, yields a net-negative return on investment.

The following table ranks the modeling approaches based on the "claim-gained per engineering-day" metric, explicitly noting their feasibility within the 16 GB RAM and six-hour constraints.

Rank	Modeling Approach	Fits in 16 GB & 6 hr?	Minimum Credible Configuration	First Point of Failure	Scientific Claim Gained vs. CAMS/Surrogates	ROI (Claim per Eng-Day)
1	1D Column Model (WRF-SCM or PySDM) driven by SMOG-India	Yes	35 vertical levels, $\Delta t = 60s$, KPP chemistry, localized surface heat fluxes.	Maintenance of accurate subsidence and lateral advective forcing datasets.	Accurately resolves highly non-linear nocturnal boundary layer titration and out-of-distribution emission responses.	High. Requires minimal setup days; yields highly localized, publishable boundary-layer chemistry insights.

2	OD Box Model (AtChem 2 / PyBox) driven by EDGAR v8.1	Yes	0D well-mixed reactor, full Master Chemical Mechanism (17,000+ reactions).	Complete loss of physical realism (no dilution, boundary layer dynamics, or mixing).	Exact identification of specific VOC-oxidation pathways and radical budgets leading to Delhi smog events.	Medium. Trivial to configure; however, claims are restricted strictly to theoretical chemical kinetic potential, not spatial forecasting.
3	WRF-Chem (3D Eulerian Online CTM)	No	10 km grid, 35 levels, MOZART/MOSAIC mechanisms.	RAM exhaustion during single-thread source compilation; 6-hour timeout during numerical integration.	N/A (Cannot run successfully).	Zero. Endless engineering days attempting to bypass hardware constraints with no usable output.
4	CMAQ / CAMx (3D Eulerian Offline CTM)	No	10 km grid, 35 levels, CB6 gas-phase mechanism	Ephemeral storage exhaustion when downloading 72h of 3D	N/A (Cannot run successfully).	Zero. Massive I/O bottlenecks prevent execution

			m.	meteorological fields.		entirely.
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The honest, empirical answer for all three-dimensional Eulerian CTMs (WRF-Chem, CMAQ, CAMx) is that they are explicitly not feasible under the prescribed CI/CD constraints. The storage bottleneck prevents offline models from acquiring necessary meteorological data, and the compute and memory bottlenecks prevent online models from compiling and solving stiff differential equations within six hours. The only scientifically valid and mathematically defensible pathway within the GitHub Actions runner is to deploy a one-dimensional Single-Column Model (such as PySDM or WRF-SCM) paired with the SMOG-India or EDGAR v8.1 anthropogenic emissions inventory.

6. Infrastructure Alternatives: Bypassing the Constraints

To change the answer from "not feasible" to "feasible" for a full 3D WRF-Chem simulation over the Delhi NCR, the underlying computational architecture must be fundamentally upgraded. While a high-performance computing MPI cluster is the standard for operational agencies, a single, powerful cloud instance can suffice for a minimal nested configuration (e.g., outer domain at 27 km, nesting down to 9 km, and an inner domain of 3 km consisting of 100 by 100 grid cells with 35 vertical levels).

To complete a 72-hour high-resolution WRF-Chem forecast in under six hours without memory allocation failures, the system requires high processor clock speeds, a minimum of 32 physical cores to distribute the computational load of the KPP chemistry solver, and at least 64 GB of RAM to handle the massive four-dimensional state vector allocations.

The smallest cloud instance that reliably changes the feasibility answer is the Amazon Web Services (AWS) **c6i.8xlarge** or **c6a.8xlarge** (Compute Optimized) instance.

- **Architectural Specifications:** These instances provide 32 vCPUs, 64 GB of RAM, and up to 12.5 Gbps of network bandwidth⁴⁸. The c6i series utilizes Intel Xeon Scalable processors, while the c6a series utilizes AMD EPYC processors. Both architectures feature advanced vector extensions highly optimized for the floating-point matrix operations inherent in Fortran-based atmospheric codes⁴⁸.
- **Cost Analysis (Asia Pacific - Mumbai / ap-south-1):** The On-Demand pricing for the c6i.8xlarge is approximately \$1.36 per hour⁴⁸. However, for scheduled daily forecasts where minor interruptions are tolerable, AWS Spot instances provide massive cost reductions, dropping the price to approximately \$0.38 per hour⁴⁹.
- **Operational Expense:** A 72-hour forecast running for roughly four hours of wall-clock time on a Spot instance would cost approximately \$1.52 per daily run. This translates to an operational expense of roughly \$45 per month.

By migrating the CI/CD pipeline to dispatch workflow jobs to a self-hosted ephemeral AWS EC2

c6i.8xlarge runner, an engineering team can execute a legitimate, high-resolution 3D WRF-Chem forecast for Delhi, entirely bypassing the crippling limitations of the free-tier hardware while maintaining an automated, scheduled pipeline.

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